PATHOPHYSIOLOGY — NURSING

Immune System — Part 1

NRSG 224 — Recorded lecture 4. Exam-detail notes from the recording. Slide bullets plus what she said that never appeared on the slide. Red text is added from her posted study-guide handout.

She said (not on the slide): *This is a pre-class lecture. The immune system is complicated enough to fill a quarter; she is compressing it into two class periods and wants a few things remembered rather than a long list forgotten.*

1. Four functions of the immune system

She collapsed the job list to four. The first is the one people already know. The other three are the ones students skip.

Function	What she wants you to say
Immunity	Resistance to disease by defending against pathogenic organisms.
Self vs non-self	Recognize foreign molecules and get rid of them.
Inflammation	Not just a bad feeling. It is a tool to defend against harmful agents and start healing.
Cancer surveillance	Callback to the cancer lecture: abnormal cells probably arise all the time; the immune system is supposed to recognize and clear them.

She said (not on the slide): *These are not four separate machines. The same self-vs-non-self mechanism is used to spot a cancer cell and to spot a pathogen. Lots of overlap.*

Self vs non-self works because almost every cell has surface antigens with unique molecular shapes. White cells have learned which shapes belong to you. Her handout name for those shapes is antigenic determinant sites, or epitopes. She contrasts autoantigens (normal body markers) with foreign antigens.

Antigen vs antibody — she stopped the lecture for this

DO NOT MIX UP: Antigens and antibodies are the two words she sees mixed up most often on the test.

- Antigen = a marker on a cell membrane used to decide self vs non-self. Antigens are not automatically bad. A self antigen is left alone. A foreign antigen is targeted.
- Antibody = what the immune system MAKES to get rid of an antigen it has decided is foreign.

TEST TIP: Antigen in and of itself is neither good nor bad. The interpretation (self vs foreign) decides.

2. Two components of the immune system

Immune cells

Fancy name for white blood cells, plus the substances they make: antibodies and cytokines. She pointed back to the blood lecture for the cell types.

Lymph structures — primary vs secondary

Kind	Organs she named	Job	If it fails
Primary	Bone marrow and thymus	Where immune cells are produced or mature	Impairs immunity a lot, especially bone marrow
Secondary	Spleen, lymph nodes, tonsils, appendix, patches in GI, respiratory, and reproductive tracts Her handout names the GI patches Peyer's patches, and the mucosal tissue MALT (mucosa-associated lymphoid tissue) in respiratory, GI, and reproductive systems. Role she wrote: trap antigens and promote their interaction with mature immune cells.	Clean foreign material out of lymph before it returns to blood	Less efficient, but people lose tonsils or appendix all the time and do not get sick constantly

TEST TIP: Primary = born/matured there. Secondary = filters. Losing a secondary organ is not the same as losing bone marrow.

3. Two major immune mechanisms

	Innate	Adaptive
Also called	Natural / native / nonspecific	Specific / acquired
When ready	Present at birth, jumps in right away	Acquired later; not ready at birth
How picky	Anything foreign. Not specific.	Learns the particular pathogens you meet
Speed vs power	First line. Holds pathogens until adaptive kicks in, and helps activate adaptive.	Slower to start, more effective because it is aimed at one antigen

She said (not on the slide): *Together they are a complete package: innate covers you immediately against anything foreign; adaptive takes longer but does not spread its attention thin.*

4. Five components of innate immunity

1. Skin and mucous membranes

She said (not on the slide): *Intact barriers keep pathogens out. Preventing infection is more efficient than clearing one that already started. A break in the skin loses this first line.*

2. Phagocytic white cells

Mainly neutrophils and monocyte-macrophages. Two jobs:

- Engulf and digest foreign particles, tumor cells, and virus-infected cells, holding the line until adaptive immunity starts.
- Act as antigen-presenting cells (APCs). They process the foreign antigen and display it on their membrane so a T cell can see it. That display is what activates adaptive immunity.

3. Natural killer (NK) cells

She said (not on the slide): *NK cells are the only lymphocyte in innate immunity. T cells and B cells are adaptive.*

Job: kill tumor cells and virus-infected cells. Not by phagocytosis. They release toxins or enzymes that damage the target.

4. Cytokines

Expanded definition: a hormone-like protein released by certain white cells that acts on other white cells to regulate the immune response. A white cell telling other white cells what to do.

Examples she named: interferons, interleukins, tumor necrosis factor, colony-stimulating factor.

What they can do (her study-guide list is examples, not complete):

- Increase or decrease production, maturation, and activity of white cells (CSF / growth factors).
- Mediate inflammation.
- Produce fever. Handout mechanism: cytokines act on the hypothalamic thermoregulatory center and raise the set point.

She said (not on the slide): *Fever is useful. Many pathogens cannot survive the higher temperature, and fever makes some white cells more active. Some cytokines do more than one job; functions overlap. Clinical reason she cares: we now give synthetic cytokines, and we have drugs that block cytokines. If you know the job, you can figure out why the drug is ordered.*

5. Complement system

She said (not on the slide): *She compared it to the clotting cascade: both are proteins sitting in blood in inactive form; each factor activates the next. End of clotting = clot. End of complement = a more efficient immune system, plus inflammation.* Her handout: a cascade of about 20 plasma proteins.

Job: complement the work of phagocytes and antibodies in clearing pathogens.

TEST TIP: If she lets you remember only one word with complement, she said that word is inflammation.

What activation does	Spoken meaning
Mast cells and basophils degranulate	Mast cells are very similar to basophils. Granules hold mediators of inflammation: histamine, leukotrienes, bradykinin. Degranulation = inflammation.
Enhanced chemotaxis	Chemical signal that calls white cells to the area that needs them.
Lysis	Bursts pathogens or abnormal cells, including cells that have become cancerous.
Opsonization	Coats antigen-antibody complexes with complement pieces (green balls on her slide) so phagocytes can grab them. Many bacteria have a capsule that is hard to phagocytize. Antibody gives a handle; complement makes the handle even easier to pull in.

5. Adaptive immunity — T cells and B cells

Provided by T and B lymphocytes. Each T or B cell is programmed for ONE specific foreign antigen. That is the difference from innate.

Branch	Cell	Targets	How
Cell-mediated	T cells	Microbes INSIDE cells — usually viruses	Cannot enter the cell either, so it kills the infected cell
Humoral	B cells	Microbes OUTSIDE cells — usually bacteria	Antibodies. Antibodies cannot go inside cells, so this branch is for extracellular targets

She said (not on the slide): *Bacteria can sometimes go inside cells. If they do, T cells take that job too.*

Helper T (CD4) — the master switch

CD4 is the membrane marker used to identify this T cell. Almost all white cells have markers; CD4 is the helper's.

Called helper because, once an APC shows it the antigen it was programmed for, it releases cytokines that activate the other T cells and the B cells.

She said (not on the slide): *The diagram also shows it influencing macrophages, neutrophils, and dendritic cells. It does NOT need to activate innate immunity. It DOES need to activate the rest of adaptive immunity. It still makes innate cells work more efficiently. Wipe out helper T and adaptive immunity does not turn on, and innate immunity works worse. That is why she calls it the master switch.*

Cytotoxic / killer T (CD8)

Kills tumor cells and virus-infected cells by releasing toxins and enzymes — same method as NK cells. Two differences she made you name:

- Cytotoxic T is adaptive, so it must be activated by helper T first. NK is innate and does not need helper T.
- Cytotoxic T targets one specific foreign antigen. NK attacks anything foreign.

Suppressor / regulatory T

Two jobs:

- Slow or stop the immune response once the pathogen is gone. Checks and balances so the response does not run on.
- Suppress self-reactive white cells (cells that would attack normal self). That is how it helps prevent autoimmune disease. Her handout adds that regulatory/suppressor T cells are believed to be a subset of CD4+ or CD8+ cells.

Memory T

Hangs around a long time and remembers that pathogen. You do not need a large standing army of helper and cytotoxic T cells for an infection that is already gone, but a second exposure should be recognized fast enough to mobilize before you get sick.

She said (not on the slide): *She read that memory T cells may live longer because they have more mitochondria, so they can make more ATP. Cute cartoon: the memory cell telling E. coli "I'll never forget you" means next time I will kill you.*

Humoral side — plasma cells and memory B

B cells also must be activated by helper T first, then specialize.

- Plasma cells make antibodies. Antibodies bind surface antigens and act like handles so phagocytes can grab encapsulated bacteria. Antigen + antibody = antigen-antibody complex. Coat that with complement (opsonization) and clearance is even easier.
- Memory B hangs around. On re-exposure it quickly drives proliferation of plasma cells so a large antibody burst happens before you get sick the second time.

6. Five antibody classes

Class	Slide line	What she added
IgG	Most plentiful; crosses placenta	Only one that crosses the placenta. Gives newborns most of their protection for the first few months.
IgA	Secretory IgA; mucous membranes	Found in tears, saliva, GI and respiratory secretions. Those secretions protect mucous membranes, which are portals of entry. Genetic inability to make IgA → eye, respiratory, and GI infections.
IgM	First to appear; suggests current infection	Does not hang around. Hepatitis example: IgM against the virus = recent or current. IgG against the virus = they had it sometime (2 months, 2 years, or 10 years — you cannot tell).
IgD	On the B cell; stimulates proliferation	Not usually free in blood. Sits on the B-cell membrane like a monitor. If its antigen is still around and binds, the B cell proliferates and makes more antibody.
IgE	Allergic reactions; defends against parasites	In this country we mostly remember the allergy half because parasite disease is uncommon. She postponed the allergy detail to next class.

TEST TIP: IgM = now. IgG = sometime, and it lasts. IgA = secretions / mucosa. IgG is the placental one. IgE = allergy and parasites.

7. Active vs passive immunity

She said (not on the slide): *Students mix these on the test. Her rule: active means YOUR immune system had to generate it. Passive means your immune system did nothing and just received it.* Her handout: both active and passive are types of adaptive immunity, not innate. Active is long-lasting but takes a few days to weeks after the first exposure. Passive is short-term only, and can be antibodies or immune cells received from another source.

Example she walked	Which	Why
You get strep throat and make antibodies	Active	Your system made them
You get a vaccine	Active	The shot is antigen. You still have to make the antibody.
Mom's antibodies cross the placenta to the fetus	Passive	Mom made them, not the baby
Gamma globulin injection after a hep B needle stick	Passive	Somebody else made the antibody; employee health injects it

Slide line to memorize: gamma globulin (IgG) = antibody → passive immunity. Vaccine = antigen → active immunity.

She said (not on the slide): *People mix them because both can be a shot. The difference is what is in the syringe.*

	Passive	Active
Starts	Right away — protected as soon as the antibody arrives	Slow. Antibodies have to build up. COVID vaccine example: not protected until about 2 weeks later.
Lasts	Not long. Antibodies break down. Your system does not know how to replace them because it did not make them.	Long. When those antibodies age out, your system knows how to make more.

She said (not on the slide): *The summary chart on the next slide is reused in third quarter adult health 1. She told you not to cram and dump it.*

8. Immunodeficiency disorders

One or more branches of the immune system are not working.

Expected problems:

- Lots of infections: persistent, hard to treat, or caused by organisms that usually do not bother people.
- Higher cancer incidence, because cancer surveillance fails.

She said (not on the slide): *Severity depends on how many branches are hit. It can be one antibody class, all antibodies, B and T both, or only phagocytes, or only complement.* Her handout's five deficiency types: B-cell immunodeficiency; T-cell immunodeficiency; combined T-cell and B-cell immunodeficiency; disorders of the complement system (one or more components missing); disorders of phagocytosis (neutrophils, monocytes/macrophages).

Phagocyte failure patterns she listed:

- Can engulf but missing the enzymes to kill what was ingested.
- Can kill but lost chemotaxis, so they do not know where to go.
- Know where to go but lost mobility, so they cannot get there.

Complement failure: missing an activating factor → cannot generate inflammation, so one defense tool is gone. Missing an inhibiting factor → inflammation cannot be slowed once it starts.

Primary vs secondary

	Primary	Secondary
When	Born with it	Acquired later
Why	Inherited gene defect / mutation → defective white cells that cannot do the job	Infection (AIDS), cancer (leukemia), immunosuppressive drugs, or another organ failure that hits immunity Her handout's cancer example is lymphoma, and she also lists other diseases that affect synthesis or maintenance of immune cells or components.

She said (not on the slide): *Liver makes complement factors, so liver failure compromises complement. Some kidney diseases damage the glomerular membrane so antibodies, normally too big to leak, spill into urine and are not in the blood where they protect you.*

9. AIDS — the immunodeficiency she chose

DO NOT MIX UP: AIDS is an immunodeficiency disorder, not an autoimmune disorder. It is in the name: Acquired Immunodeficiency Syndrome. She said people say the acronym and then forget that.

Cause: HIV, human immunodeficiency virus. A retrovirus that selectively attacks the immune system, mainly helper T cells.

Lose helper T and you lose all adaptive immunity (no master switch) and innate immunity works less efficiently.

Opportunistic infection = infection by an organism that does not bother people with a normal immune system.

How HIV replicates

She said (not on the slide): *Retrovirus has RNA only, not DNA. Most viruses have one or the other, not both.*

- Enters a host cell, usually a helper T.
- Changes viral RNA into viral DNA. The enzyme on her handout is reverse transcriptase.
- Inserts that viral DNA into the host cell's DNA.
- Host DNA is the protein recipe, so the cell now makes more virus.
- Virus numbers rise until the helper T breaks apart and new HIV is released to infect other helper T cells.

Early on the body replaces many killed helper T cells and kills many viruses. Over time helper T counts fall, virus rises, and immunity fails.

Three routes of transmission

Route	What she actually said
Blood / blood products	Not mainly transfusion. U.S. blood has been screened since the mid-1980s. A middle-of-the-road figure she used: about 1 in 2 million transfusions. She also cited about 4.5 million Americans a year who would die without a needed transfusion — so if you need blood, have it. Real blood route: shared needles among IV drug users. Healthcare needle sticks are rare but have transmitted it. Be careful with needles.
Sexual contact	Semen and vaginal secretions. Not casual contact: not a handshake, toilet seat, or shared glass.
Perinatally	Usually NOT in utero — placenta is a good barrier. Usually exposure to mom's blood in labor and delivery, or breastfeeding. Depends on mom's viral load.

Two stages

Stage	Definition she used
HIV infection	Infected, no immunodeficiency symptoms yet. Usually no symptoms until helper T counts are very low. Latent: virus is there, immune system keeps viral levels low. Can last up to 10 years or more even without antivirals; drugs may stretch it longer. Dangerous because they can still transmit. Her handout: in the latent phase the cell serves as a reservoir for

	the virus.
AIDS	Diagnosis switches when helper T count drops below 200, OR they develop an AIDS-defining illness. Normal helper T she gave: about 600 to 1,200.

TEST TIP: AIDS threshold: CD4 below 200, or an AIDS-defining illness. Normal frame: 600–1,200.

Window period

Time between infection and seroconversion (blood test flips from negative to positive). During it, nobody knows, so nobody is preventing transmission. That is the dangerous part of the latent phase.

She said (not on the slide): *Length depends on the test. Textbook: 1 to 3 months with the cheaper screening tests used on large numbers of people. Blood banks use a more expensive test that shortens the window to about 2 or 3 weeks. Many hospital patients may not know they have HIV, so use standard precautions on everybody. She said this is not because patients lie; they may not know.*

Tests

Test	Use
EIA (enzyme immunoassay; also called ELISA / enzyme-linked immunosorbent assay)	Common screen. Not HIV-specific — a technique also used for other viruses (she mentioned CMV). Has false positives, so a positive must be confirmed. She prefers false positives over false negatives on a screen so infected people do not fall through.
Western blot	Follow-up if EIA is positive. More expensive, more specific. Positive = HIV infection. Negative = the EIA was a false positive.
Home finger-stick mail-in	Lab runs both tests. Results by phone in about 3 to 7 days.
Home oral swab	You read it in about 20 minutes. EIA only. A positive still needs a Western blot.

Once it is AIDS

CDC AIDS surveillance case definition is a list of AIDS-defining illnesses, mostly opportunistic infections and cancers. Infections become more numerous, more severe, and harder to treat — both ordinary ones and opportunistic ones.

She said (not on the slide): *Without antivirals, death in about 2 to 3 years. Antivirals extend that.*

Assessment findings she reviewed

System	What to know
Respiratory	Pneumocystis pneumonia is the most common AIDS pneumonia. Fungus: Pneumocystis jirovecii (on the slide). Old name Pneumocystis carinii is the animal form. TB (Mycobacterium tuberculosis) also occurs and is NOT opportunistic. Related opportunistic cousin: Mycobacterium avium complex — same genus, different species, TB-like symptoms, not as hardy, so it hits impaired immunity. Ordinary bugs still happen: Streptococcus pneumoniae, Haemophilus influenzae.
GI — mouth and esophagus	Anorexia, nausea, vomiting. Lesions from Candida, CMV, herpes simplex. Painful eating and swallowing. Retrosternal pain = behind the sternum, where the esophagus runs. Epstein-Barr virus can cause oral hairy leukoplakia.
GI — gastroenteritis	Inflammation of stomach and intestines. Primary symptom is diarrhea because inflamed gut cannot absorb fluid. In AIDS it can last months, antibiotics often fail, large fluid and nutrient losses, weight loss.
Nervous system	Toxoplasmosis, Cryptococcus, herpes. Meningitis (meninges) or peripheral neuropathy.

	Brain tumors also occur because surveillance is down.
HAND	HIV-associated neurocognitive disorder. Maybe half of AIDS patients. Diagram: most have no symptoms or only mild ones. Only a small percent progress to dementia. Most late-stage patients are not confused. Oriented, maybe a little lethargic or slow, still in touch with reality.

Cancers

- Kaposi sarcoma: a vascular malignancy — cancer of cells lining some smaller blood vessels. Raised reddish-purple skin lesions, often trunk or head and neck, can be on internal organs and then cause organ dysfunction. Looks different person to person. Head and neck lesions disfigure and hurt.
- Cervical cancer and non-Hodgkin lymphoma also occur. Neither is opportunistic; both can happen with a normal immune system too.

Wasting syndrome

She said (not on the slide): *Skinny / wasting away is not enough. To count as wasting syndrome the weight loss has to happen WITHOUT an opportunistic infection or cancer that could explain it. In parts of Africa with a lot of AIDS it is called slim disease because wasting is so common.*

Interventions — two levels

- Treat the infection or cancer that already happened with its standard therapy: antibiotics, antifungals, chemo.
- Treat the underlying HIV with antivirals that slow replication — block entry into new cells, or interfere with enzymes the virus needs to copy itself.

She said (not on the slide): *Drugs have slowed progression and improved survival enough that AIDS is now treated as a chronic illness. New drugs now raise questions that did not matter when survival was short, such as whether a drug raises cholesterol and heart-disease risk. Undetectable viral level is not the same as cured. We cannot yet remove the viral DNA sitting in host-cell DNA, and that DNA can still make new virus. She will not say we have cured AIDS.*

10. Rapid-fire test bank

- Four functions: immunity, self vs non-self, inflammation, cancer surveillance. Overlap is expected.
- Antigen = marker, not automatically bad. Antibody = what we make against a foreign antigen.
- Primary lymph organs: bone marrow and thymus. Secondary: spleen, nodes, tonsils, appendix, mucosal patches. Primary failure hurts more.
- Innate = born with it, nonspecific, fast. Adaptive = acquired, specific, slower, stronger.
- Five innate pieces: barriers, phagocytes/APCs, NK cells, cytokines, complement.
- NK is the only innate lymphocyte. Kills with toxins, not phagocytosis.
- Cytokine = white cell telling other white cells what to do. Fever is useful. We give synthetic cytokines and cytokine blockers.
- Complement word = inflammation. Also chemotaxis, lysis, opsonization. Mast cell ≈ basophil; histamine, leukotrienes, bradykinin.
- B cells / antibodies = outside the cell, usually bacteria. T cells = inside the cell, usually viruses, by killing the infected cell.

- CD4 helper = master switch. CD8 cytotoxic needs helper T and is antigen-specific; NK does not and is not. Regulatory T stops the response and stops self-reactive cells. Memory T remembers; maybe more mitochondria.
- Plasma cell = antibody. Memory B = fast second response.
- IgG placenta and most plentiful. IgA secretions. IgM now. IgD on B cell. IgE allergy and parasites.
- Vaccine = antigen = active. Gamma globulin = antibody = passive. Passive is fast and short. Active is slow and long. COVID example: about 2 weeks.
- Immunodeficiency: infections plus cancer. Primary = born with it. Secondary = AIDS, leukemia, immunosuppressants, liver (complement), kidney (antibody loss in urine).
- AIDS is immunodeficiency, not autoimmune. HIV hits CD4. Retrovirus: RNA → DNA → into host DNA.
- Routes: shared needles more than transfusion (~1 in 2 million). Sex, not casual contact. Perinatal usually at delivery or breastfeeding, not in utero.
- Window: 1–3 months on cheap screens; 2–3 weeks at the blood bank. Standard precautions on everyone.
- EIA screen, Western blot confirms. Home oral swab is EIA only.
- AIDS: CD4 < 200 (normal 600–1,200) or AIDS-defining illness. Without antivirals, about 2–3 years.
- PCP = P. jirovecii, most common AIDS pneumonia. MAC is opportunistic; TB itself is not. HAND rarely reaches dementia. Wasting syndrome has no explaining infection or cancer. Undetectable is not cured.

11. End of deck

Slides 85–93 are cartoons with no recorded audio. She closed with: enjoy the cartoons, see you in class.